How To Become a Physics Pro

Proposed by:

Team Curiosity

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Problem Statement

Let's be for real: Physics can be one of the most challenging subjects for Grade 8 students. Don't deny it, many students struggle to understand concepts such as ***motion, force, speed, work, energy, gravitational laws, and heat transfer*** because the topics are way too complicated and require the strongest abstract and critical reasoning.

Did you know: Students often rely on memorisation (stock-knowledge) rather than dedicated understanding. When dozens of formulas, units, and scientific terms accumulate, scholars lose confidence and stimulus. This results in low scores, frustration, and a weaker appreciation for how fun physics can be.

We interviewed **Katara, an 8th grader from 8-Camia**, who communicated that she finds physics "too heavy" and wishes for a study method that explains the concepts thoroughly, makes learning interactive, and helps her review without stress. Her very experiences are what grade 8 students face, and let's be real, that summative assessment exam for the grade 8's would really be an entire list piled up all together.

Create a software that can be a viable study method for students, whether they use a premade quiz for a topic or they make their own quiz to help others! Students often struggle to find effective and interactive study methods tailored to their individual learning styles and their own academic needs. Pressure builds up when the entire school year's topic of numerous subjects is going to be the final decision maker of whether you will be a Director's Lister.

Through this project, we can create an engaging way of studying or a collaborative way of learning for Physics. This is the project **"Gravity pulls with Curiosity."** Where limited ideas are taken into software with no limits.

Project Objectives

Pressure builds up: When faced with an exam that shows your efforts, you don't want to fail it, duh! I, Edgar, am a Physics geek. I want to show my peers how Physics can be an engaging subject, for them to understand why I cherish it so much. I, Viktoria, am a compassionate person. I want to embody kindness and to show it; this project is for me.
I, Seth Bito, a lover man, with loyalty, we can show that challenges are a step back. This is how we can give back to our peers, to firstly eliminate their fears.
But let's be honest: Memorising thousands of terms can feel overwhelming, it turns us into a dictionary that doesn't even have definitions, which is why
This project aims to:

1. Provide a platform for **Physics** quizzes focused on Grade 8 topics, such as:
    - Ideas of Motion
    - Kinematic Quantities
    - Newton's Law of Motion
    - Force
    - Impulse and Momentum
    - Conservation of Momentum

- Historical Development of the Universal Law of Gravitation
- Gravitational Force, Field, and Potential Energy
- Acceleration due to Gravity
- Conservation of Mechanical Energy
- Heat Transfer
- Consequences of Heat Transfer
- Climate Change and Renewable Energy
- Introduction to Waves
- Characteristic Behaviors of Waves
- Sound as a Wave and the Physics of Echolocation
- Historical Development of Ideas about Light
- Colors
- Wave Nature of Light
- Electromagnetic Wave

2. **Improve students' comprehension** by giving feedback, hints, and corrections in real time.
    - Offer clear and well-structured explanations on the validity of the answer; this is why you were correct/wrong.
3. Make learners appreciate physics, allowing them to understand that physics can be less intimidating and easier to understand.

Actual Inputs and Outputs

User Log In System:

Output → Sign In or Sign Up

User Choices: → [1, 2]

Output → You are now logged in!

→ Try again next time.

User Interface:

Output → Banner

Welcome to Gravity Pulls With Curiosity

User Choices (Main Menu) - Input

Upon starting the program, the user will be given three choices:

1. **Physics Mission**
2. **Instructions Manual**
3. **Exits**

The user inputs any of the choices: **[1, 2, 3]**

Outputs:

- A confirmation message that the quiz has been saved
- A preview of the question has been added.
- A message notifying the user that their quiz is ready for use

If the User Chose **“Take a Quiz”**

Inputs:

The user is then asked to select a physics topic:

- Select a topic in a specific quarter

[Quarter 1: TOPICS

Quarter 2: TOPICS

Quarter 3: TOPICS]

The user is then asked to select the difficulty.

- Select Difficulty
    - Easy - 5 Questions
    - Medium - 10 Questions
    - Hard - 20 Questions
    - Limitless - The program randomizes it [20 - How many questions are there]

The user inputs answers

- For each question, the user chooses an answer by clicking (A, B, C, D, True, or False)

Outputs:

A question will be displayed on the screen, containing:

➔ **Question Text**

➔ Choices (A - D, **Multiple Choice)**

1. If the question were a multiple-choice, the user would be given four choices to answer from.
    a. If they get it correctly, the software will congratulate them, and it will also explain how the answer was correct.
    b. If they get it incorrectly, the software will motivate them and give them hints, and the software will explain why it’s wrong.

UPCOMING:

2. If the question were a modified true or false question, they would just input true if it's true; else, if it's false, they have to explain why.
    a. If they get it correctly, the question will be explained.
    b. If they get it incorrectly, the question will be explained, they are given a chance to answer again, but they just have to choose true; if they answered false (modified) or if they answered true, they have to explain why it's false.
3. If the question was true or false, the same rules from the multiple-choice test will be applied.

REVIEW PAGE

- ➔ Be able to check your data from past quizzes, where you went wrong, right, and learn from the explanations.

If the User Chose "**Menu / Instructions**"

Explains what the program does, how to use it, and menu options.

→ 📜 PROGRAM INSTRUCTIONS & MANUAL 📜

Output:

WHAT IS THIS PROGRAM?

Gravity Pulls With Curiosity is a study tool built specifically

for Grade 8 students to master Physics. It uses gamified quizzes to help you understand Motion, Force, Energy, and Heat Transfer.

HOW TO USE IT:

1. Navigation: Use the numbers [1-5] to select menu options.")

2. Quizzes: Type the letter (A, B, C, or D) of your answer.")

3. Scoring: Correct answers give full scores. Mistakes have a -0.25 penalty, but you can retry until you get it right!

Menu Meanings:

MENU OPTIONS

[1] Physics Mission: Take premade quizzes based on school quarters.

[2] Instructions: Open this manual.

[3] Exit: Safely closes the system.

Logic Plan

```
// --- GLOBAL SETUP

SET TeamMembers = ["Fortes, Edgar", "Bito, Seth", "Ancheta, Viktoria"]



// --- STARTUP ANIMATION ---

CALL function `display_robot_watermark.`

i9CALL function `loading_bar_animation`



// --- PART 1: AUTHENTICATION ---

PRINT "Authentication Required"

INPUT "Choose [1] Log In or [2] Sign Up" AS Authorization

INPUT "Username" AS username

INPUT "Password" AS password

IF authorization IS "2" THEN

    CALL `update_data` to save the new user to "users.json."

    PRINT "Account created."

ELSE

    CALL `update_data` to verify credentials from "users.json."
```

```
    IF credentials are invalid THEN
        PRINT "Login failed."
        EXIT PROGRAM
    END IF
END IF


PRINT "Welcome message from Team Curiosity."
CLEAR SCREEN
// --- Part 2: MAIN MENU ---
FUNCTION `main_menu`:
    PRINT "Team Curiosity OS v1.0."
    PRINT "[1] Launch Physics Mission"
    PRINT "[2] Menu/Instructions"
    PRINT "[3] Exit"
    INPUT "Select option" AS choice
    RETURN choice
```

```
SET user_selection = CALL `main_menu`



// --- Part 3: BRANCHING LOGIC ---

IF user_selection IS "1" THEN

    CALL `physics_mission`

ELSE IF user_selection IS "2" THEN

    CALL `show_instructions`

ELSE IF user_selection IS "3" THEN

    PRINT "Goodbye"

    EXIT

END IF

// --- FUNCTION: PHYSICS MISSION ---

FUNCTION 'physics_mission':

    PRINT "Select Quarter:



 [1] Q1 (Force),

 [2] Q2 (Energy),

 [3] Q3 (Waves) "
```

```
INPUT sub_choice

IF sub_choice IS "1" THEN SET filename = "physics_q1.csv"

ELSE SET filename = "physics_q2.csv"
```

```
OPEN filename AND LOAD rows into `questions_list.`

PRINT "Select Difficulty:

[1] Easy,

[2] Medium,

[3] Hard,

[4] Limitless"

INPUT difficulty_choice

SET question_limit BASED ON difficulty_choice (5, 10, 20, or Random)

SHUFFLE `questions_list`

SELECT first `question_limit` questions


FOR EACH `question` IN `selected_questions`:

    SET score BASED ON Difficulty Label (Easy=1, Medium=2, Hard=5)

    SET answered_correctly = FALSE


    WHILE answered_correctly IS FALSE AND score > 0:

        DISPLAY Question, Topic, and Choices (A, B, C, D)

        INPUT "Your Answer" AS user_answer
```

```
        IF user_answer matches `question.Correct` THEN.
            PRINT "Correct!"
            PRINT Explanation (if available)
            ADD score TO `total_earned.`
            SET answered_correctly = TRUE
        ELSE
            SUBTRACT 0.25 FROM score
            IF score > 0 THEN
                PRINT "Incorrect. Penalty applied. Try again!"
            ELSE
                PRINT "Out of score for this question. The correct answer was..."
                PRINT Explanation
            END IF
        END IF
    END WHILE
END FOR
```

```
    // --- ENDING ---

    PRINT "Final Score: total_earned."

    CALL `update_data` to save `total_earned` to the user's profile in "users.json."



// --- CORE FUNCTION: INSTRUCTIONS ---

FUNCTION `show_instructions`:

    PRINT "Program Manual: What it is, How to use, Scoring rules."

    WAIT FOR USER INPUT TO RETURN TO MENU

END
```

Planned Features

Through this **software**, students can venture into these features:

- **Take premade quizzes**: In this feature, we can use our provided SGs about Physics for an efficient way of making these premade quizzes.
    - If you get it **correctly**, you will get an explanation of why it is correct. This is to take your knowledge onto the spiritway of curiosity, or in simpler terms, to add to what you know.
    - If you get it **incorrectly**, you will get an explanation on why it is wrong. But it doesn't mean you can't take the question again, just like in Duolingo, the question comes back. If you get the question wrong, you will have a .25 deduction, so you can keep taking the question until it's correct! Scores can be an integer or a floating-point number.
- **Review Correct and Incorrect:** In this software, the user will be able to review what answers they got correctly and what answers they got wrong to focus on. We can also implement an AI to give definitions, why the answer was incorrect, what the correct answer is, and primarily explain!

Conclusion

In conclusion, "**Gravity pulls with Curiosity**" is an engaging software program designed to simplify and make learning easier by providing an interactive and collaborative platform for our fellow students to engage with various topics about physics through quizzes and other features. By allowing users to create and participate in quizzes, our platform aims to develop curiosity, create creativity, and exemplify critical thinking while promoting academic excellence. With features such as creating your own quizzes, taking premade quizzes, and collaborating with different people for a study session, testing the user's knowledge. This software has the potential to make learning fun, accessible, and effective for students.

By empowering users to both create and participate, the platform actively works to foster **curiosity, spark creativity, and develop critical thinking skills**, all while encouraging academic excellence. Key features, including quiz creation, access to a library of premade content, and facilitated study collaboration, position this software to make learning **fun, accessible, and profoundly effective**.

"Gravity pulls with Curiosity" can help bridge the gap between knowledge and application, contributing to the development of a more informed, engaged, and curious community. Now our students can take that Summative Achievement Exam with ease, for Physics, relieving a big burden!

Team Curiosity!

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